

FINAL PROJECT



PROJECT TASKS

- GEM5 + NVMAIN BUILD-UP (40%) → in the tutorial below
- Enable L3 last level cache in GEM5 + NVMAIN (15%)
- Config last level cache to 2-way and full-way associative cache and test performance (15%)
 - You need to run the benchmark Quicksort in 2-way and full-way.
- Modify last level cache policy based on frequency based replacement policy (15%)
- Test the performance of write back and write through policy based on 4-way associative cache with isscc_pcm(15%)
 - You need to run the benchmark multiply in write through and write back.
- Bonus (10%)
 - Design last level cache policy to reduce the energy consumption of pcm_based main memory
 - Baseline:LRU

NVMAIN+GEM5 TUTORIAL

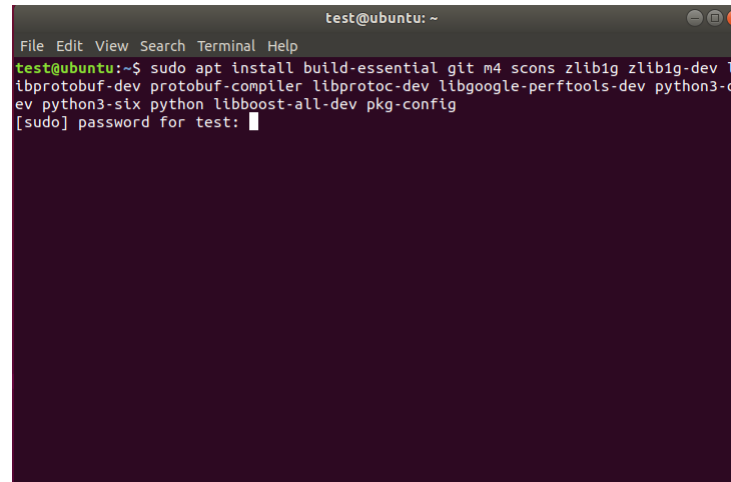


VIRTUAL MACHINE

- Please use Ubuntu 18.04
- Others settings
 - Memory space:4GB
 - Disk space :20GB

INSTALL BUILD TOOLS

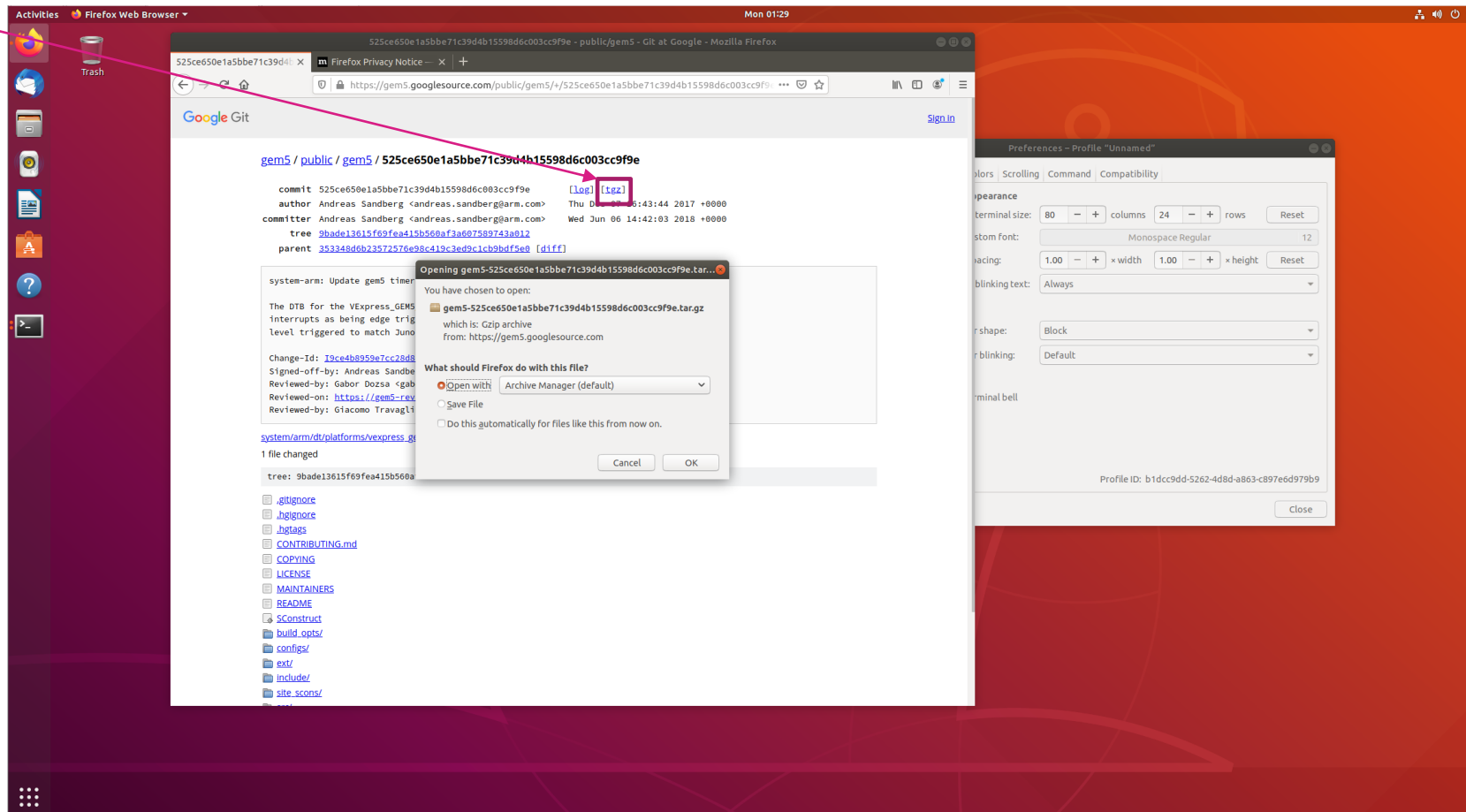
- `sudo apt install build-essential git m4 scons zlib1g zlib1g-dev libprotobuf-dev protobuf-compiler libprotoc-dev libgoogle-perftools-dev python3-dev python3-six python libboost-all-dev pkg-config`

A terminal window titled 'test@ubuntu: ~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command 'test@ubuntu:~\$ sudo apt install build-essential git m4 scons zlib1g zlib1g-dev libprotobuf-dev protobuf-compiler libprotoc-dev libgoogle-perftools-dev python3-dev python3-six python libboost-all-dev pkg-config' and the prompt '[sudo] password for test:'.

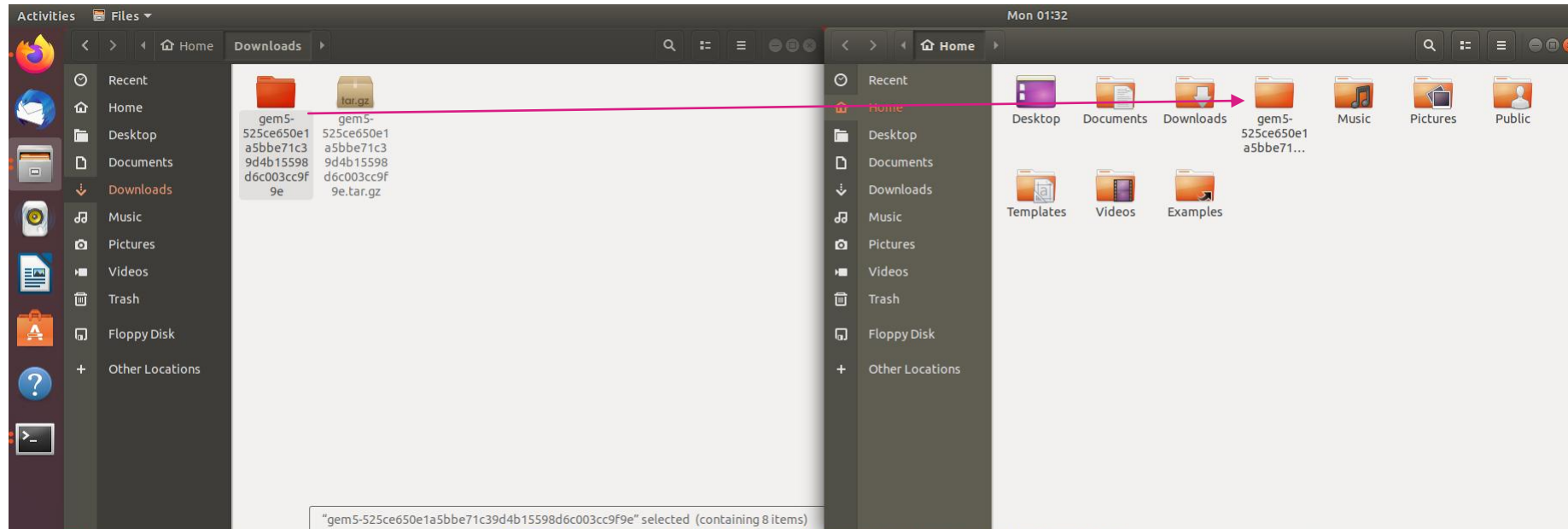
```
test@ubuntu: ~
File Edit View Search Terminal Help
test@ubuntu:~$ sudo apt install build-essential git m4 scons zlib1g zlib1g-dev l
ibprotobuf-dev protobuf-compiler libprotoc-dev libgoogle-perftools-dev python3-d
ev python3-six python libboost-all-dev pkg-config
[sudo] password for test: 
```

INSTALL GEM5

- <https://gem5.googlesource.com/public/gem5/+525ce650e1a5bbe71c39d4b15598d6c003cc9f9e>
- Press tgz Download



MOVING IT FROM DOWNLOADS TO /HOME



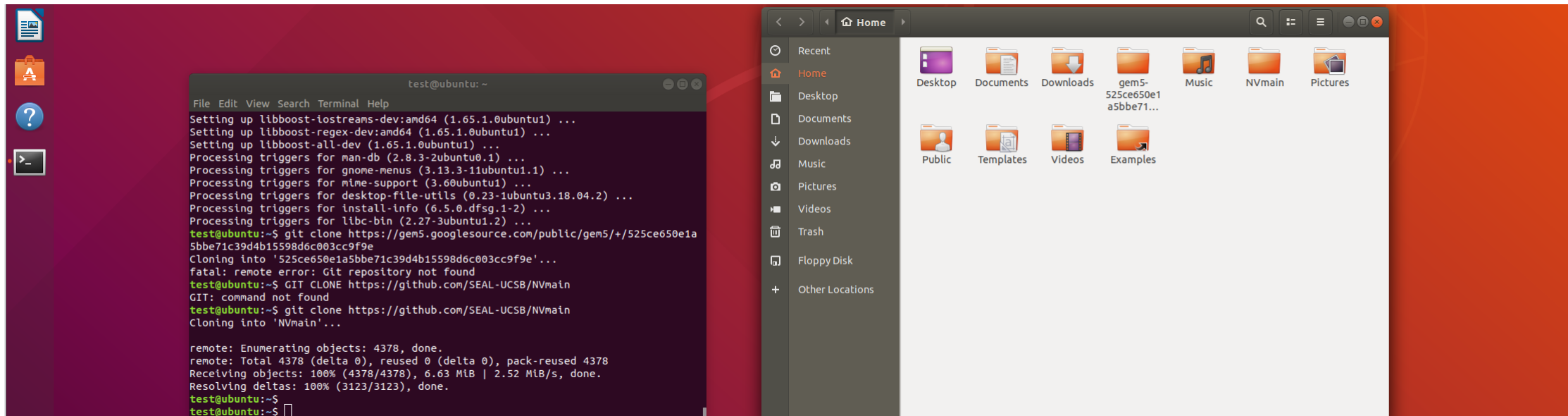
COMPILE GEM5

```
test@ubuntu: ~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e
File Edit View Search Terminal Help
test@ubuntu:~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e$ scons build/X86/gem
5.opt -j 4
```

```
test@ubuntu: ~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e
File Edit View Search Terminal Help
[ SHCC] libelf/libelf_shdr.c -> .os
[ SHCC] libelf/libelf_xlate.c -> .os
[ M4] libelf/elf_types.m4, libelf_convert.m4 -> libelf_convert.c
[ M4] libelf/elf_types.m4, libelf_fsize.m4 -> libelf_fsize.c
[ SHCC] libelf/libelf_convert.c -> .os
[ M4] libelf/elf_types.m4, libelf_msize.m4 -> libelf_msize.c
[ SHCC] libelf/libelf_fsize.c -> .os
[ SHCC] libelf/libelf_msize.c -> .os
[ SHCC] libfdt/fdt.c -> .os
[ SHCC] libfdt/fdt_ro.c -> .os
[ SHCC] libfdt/fdt_rw.c -> .os
[ SHCC] libfdt/fdt_sw.c -> .os
[ SHCC] libfdt/fdt_wip.c -> .os
[ SHCC] libfdt/fdt_empty_tree.c -> .os
[ SHCC] libfdt/fdt_strerror.c -> .os
[ AR] -> libfdt/libfdt.a
[ RANLIB] -> libfdt/libfdt.a
[ AR] -> libelf/libelf.a
[ RANLIB] -> libelf/libelf.a
[ LINK] -> X86/arch/x86/lib.o.partial
[ CXX] X86/base/date.cc -> .o
[ LINK] -> X86/gem5.opt
scons: done building targets.
test@ubuntu:~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e$
```

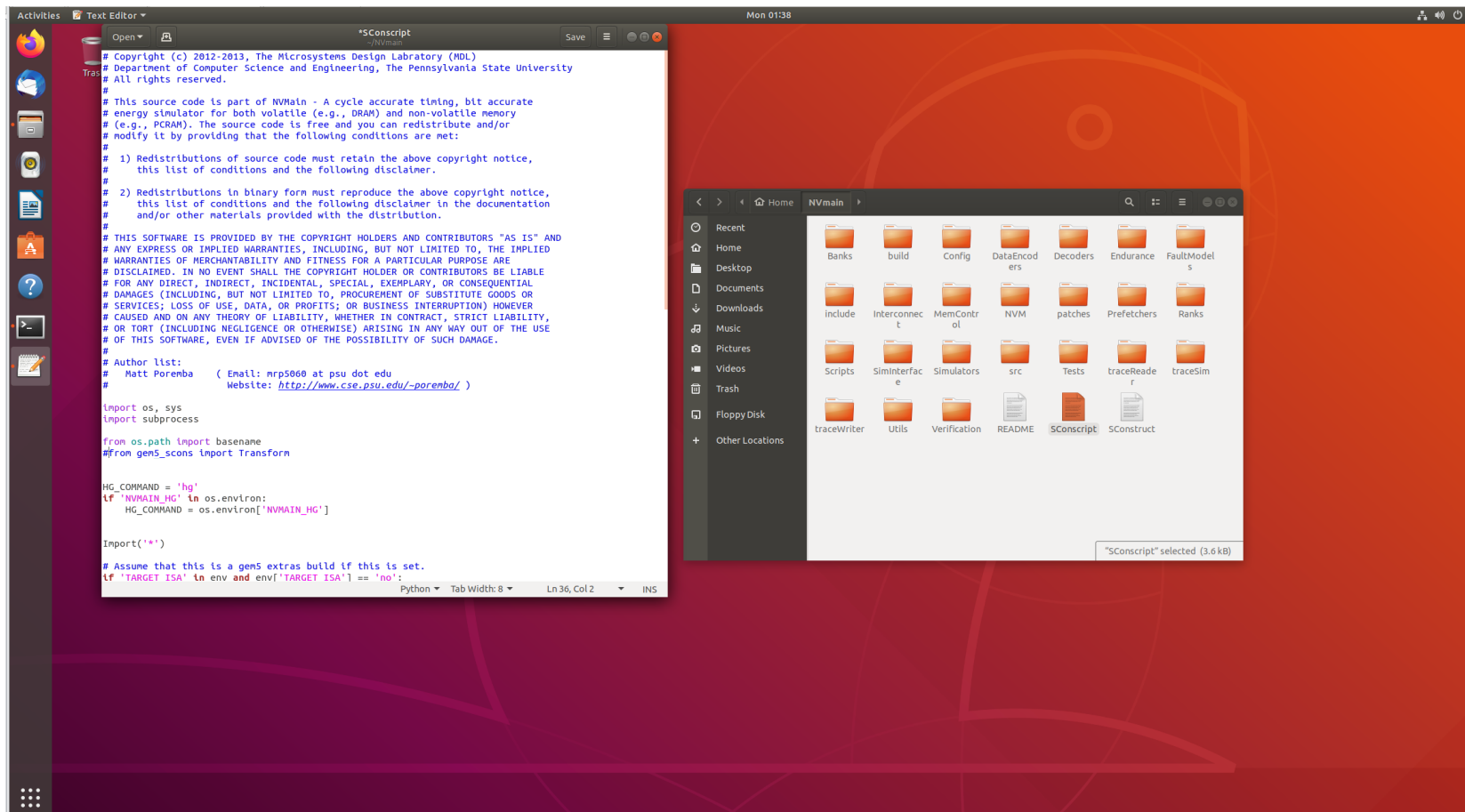

GIT INSTALL NVMAIN

- git clone <https://github.com/SEAL-UCSB/NVmain>
- Nvmain and gem5 should be in the /Home together.



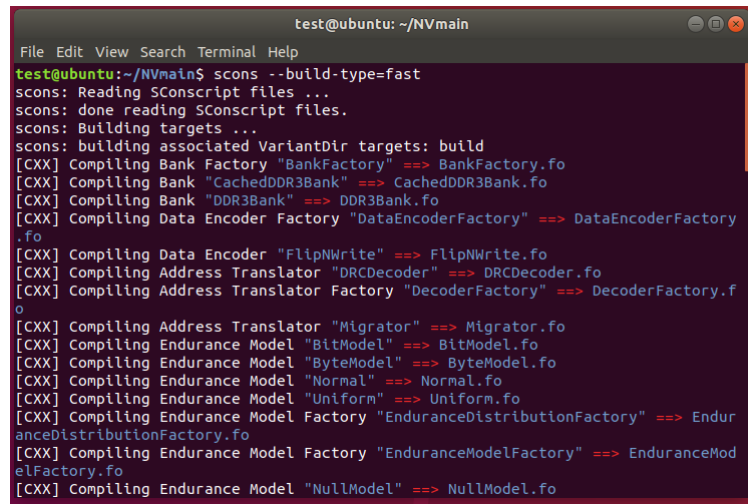
MODIFY SCONSCRIPT

- Enter the nvmain folder, click on SConscript, and annotate the 36 lines of from gem5_scons import Transform (remember to archive)



COMPILE NVMAIN

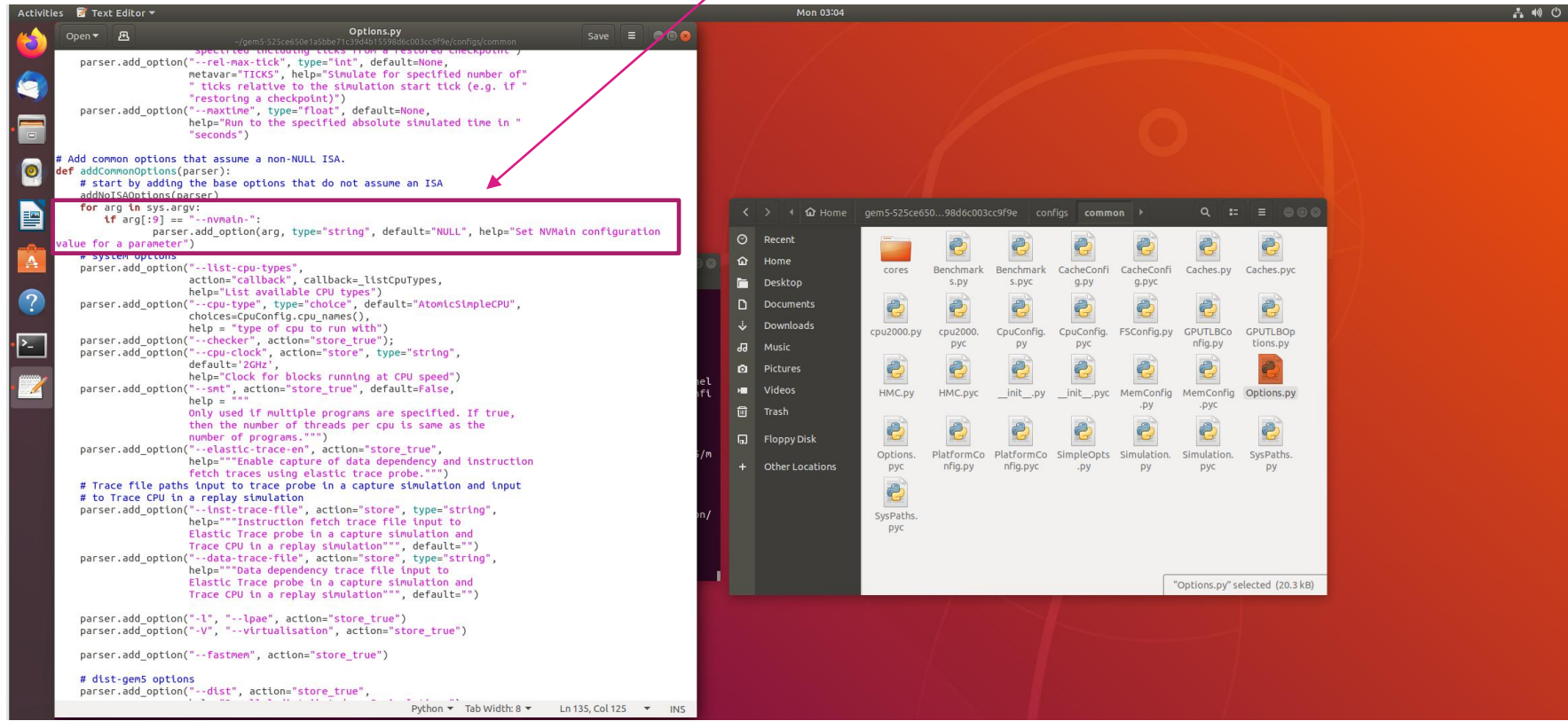
- In ./Nvmain Type: `scons --build-type=fast`



```
test@ubuntu: ~/NVmain
File Edit View Search Terminal Help
test@ubuntu:~/NVmain$ scons --build-type=fast
scons: Reading SConscript files ...
scons: done reading SConscript files.
scons: Building targets ...
scons: building associated VariantDir targets: build
[CXX] Compiling Bank Factory "BankFactory" ==> BankFactory.fo
[CXX] Compiling Bank "CachedDDR3Bank" ==> CachedDDR3Bank.fo
[CXX] Compiling Bank "DDR3Bank" ==> DDR3Bank.fo
[CXX] Compiling Data Encoder Factory "DataEncoderFactory" ==> DataEncoderFactory
.fo
[CXX] Compiling Data Encoder "FlipNWrite" ==> FlipNWrite.fo
[CXX] Compiling Address Translator "DRCDecoder" ==> DRCDecoder.fo
[CXX] Compiling Address Translator Factory "DecoderFactory" ==> DecoderFactory.f
o
[CXX] Compiling Address Translator "Migrator" ==> Migrator.fo
[CXX] Compiling Endurance Model "BitModel" ==> BitModel.fo
[CXX] Compiling Endurance Model "ByteModel" ==> ByteModel.fo
[CXX] Compiling Endurance Model "Normal" ==> Normal.fo
[CXX] Compiling Endurance Model "Uniform" ==> Uniform.fo
[CXX] Compiling Endurance Model Factory "EnduranceDistributionFactory" ==> Endur
anceDistributionFactory.fo
[CXX] Compiling Endurance Model Factory "EnduranceModelFactory" ==> EnduranceMod
elFactory.fo
[CXX] Compiling Endurance Model "NullModel" ==> NullModel.fo
```

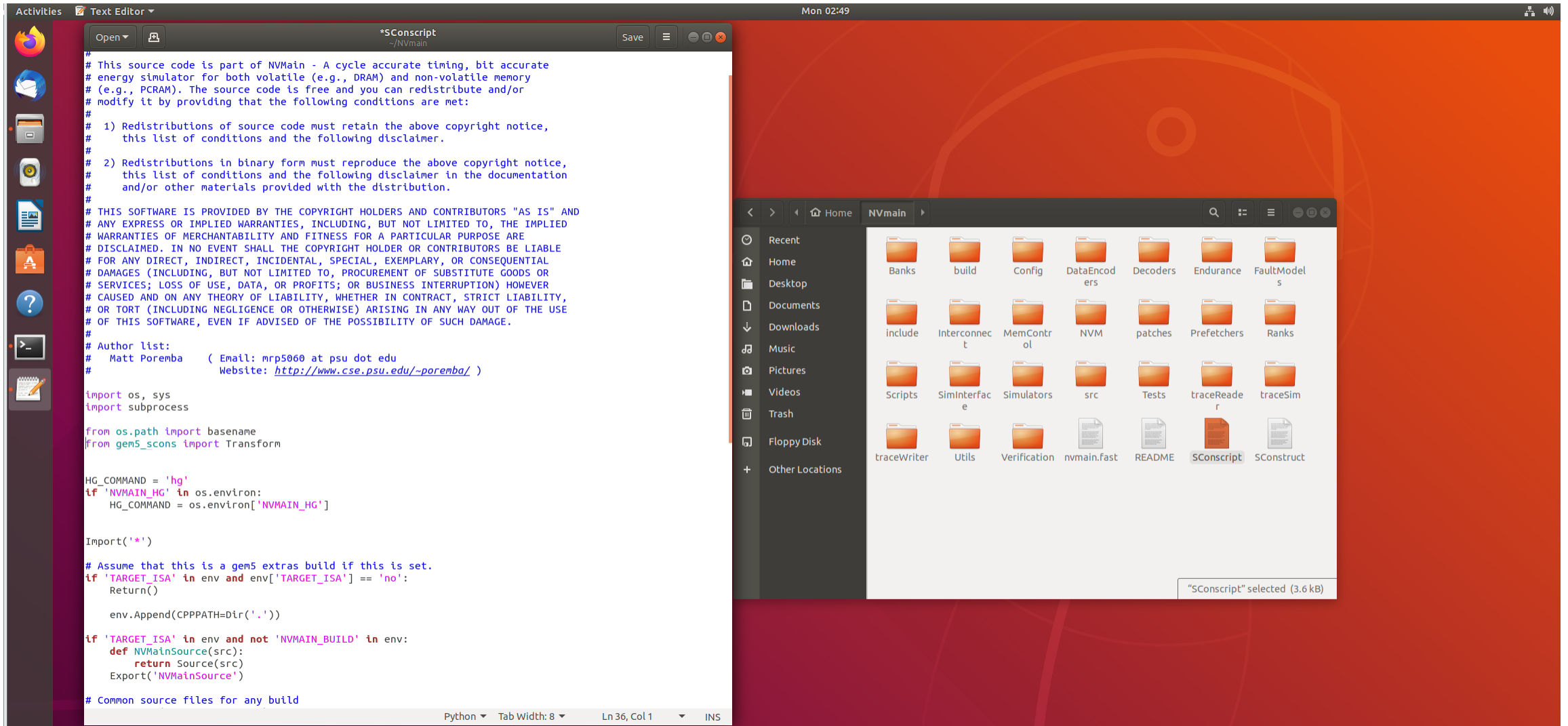
MODIFY GEM5 OPTIONS

- In gem5/configs/common/Options.py Add the next program in line 133 (remember to save it)
- for arg in sys.argv:
- if arg[:9] == "--nvmain-":
- parser.add_option(arg, type="string", default="NULL", help="Set NVMain configuration value for a parameter")



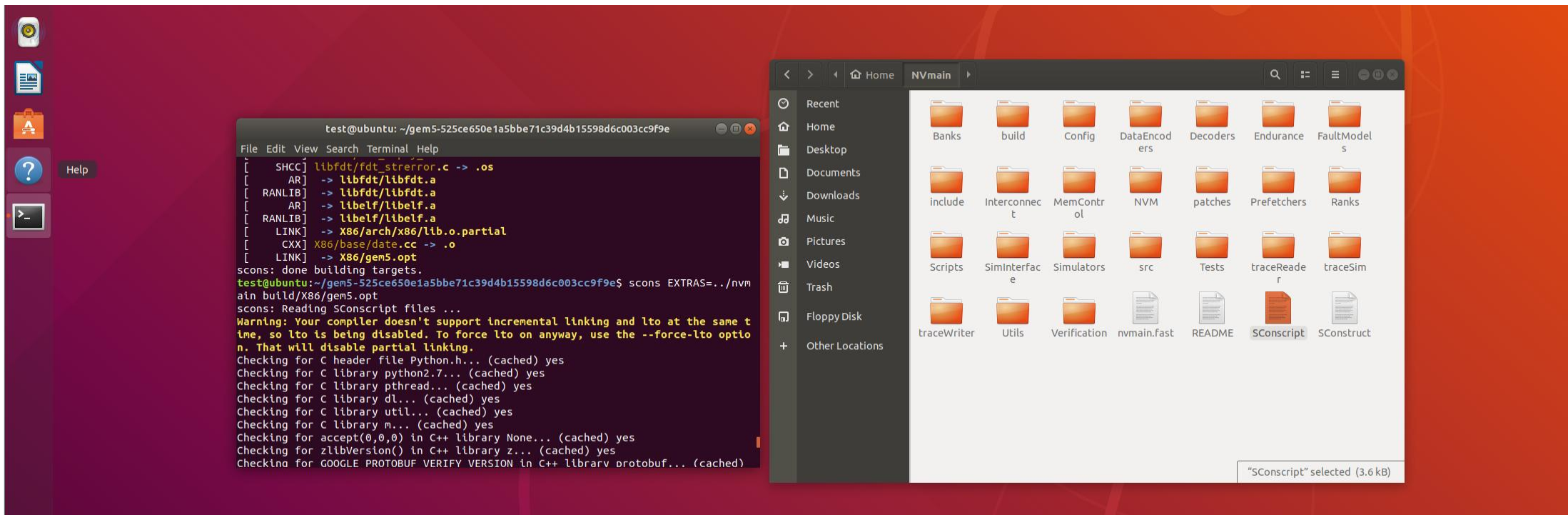
RECOVER SCONSCRIPT

- Restore the previous instructions annotated by nvmain sconscript



COMPILE GEM5

- Compile GEM5 with mixed NVMAIN in the GEM5 directory
- `scons EXTRAS=../NVmain build/X86/gem5.opt`



HELLO WORLD

Gem5

code

- `./build/X86/gem5.opt configs/example/se.py -c tests/test-progs/hello/bin/x86/linux/hello --cpu-type=TimingSimpleCPU --caches --l2cache --mem-type=NVMainMemory --nvmain-config=../NVmain/Config/PCM_ISSCC_2012_4GB.config`

nvm

```
test@ubuntu: ~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e
File Edit View Search Terminal Help
i0.defaultMemory.channel0.FRFCFS-WQF.averageTotalLatency 0
i0.defaultMemory.channel0.FRFCFS-WQF.measuredLatencies 0
i0.defaultMemory.channel0.FRFCFS-WQF.measuredQueueLatencies 0
i0.defaultMemory.channel0.FRFCFS-WQF.measuredTotalLatencies 0
i0.defaultMemory.channel0.FRFCFS-WQF.simulation_cycles 1188
i0.defaultMemory.channel0.FRFCFS-WQF.wakeupCount 0
i0.defaultMemory.totalReadRequests 0
i0.defaultMemory.totalWriteRequests 0
i0.defaultMemory.successfulPrefetches 0
i0.defaultMemory.unsuccessfulPrefetches 0
test@ubuntu:~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e$ ./build/X86/gem5.o
pt configs/example/se.py -c tests/test-progs/hello/bin/x86/linux/hello --cpu-type=TimingSimpleCPU --caches --l2cache --mem-type=NVMainMemory --nvmain-config=../NVmain/Config/PCM_ISSCC_2012_4GB.config
gem5 Simulator System. http://gem5.org
gem5 is copyrighted software; use the --copyright option for details.

gem5 compiled May 10 2021 02:54:14
gem5 started May 10 2021 03:05:52
gem5 executing on ubuntu, pid 21001
command line: ./build/X86/gem5.opt configs/example/se.py -c tests/test-progs/hello/bin/x86/linux/hello --cpu-type=TimingSimpleCPU --caches --l2cache --mem-type=NVMainMemory --nvmain-config=../NVmain/Config/PCM_ISSCC_2012_4GB.config
```

```
test@ubuntu: ~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e
File Edit View Search Terminal Help
Config: Warning: Key Decoder is not set. Using '' as the default. Please configure this value if this is wrong.
Config: Warning: Key RankType is not set. Using '' as the default. Please configure this value if this is wrong.
Creating 4 banks in all 8 devices.
Config: Warning: Key BankType is not set. Using '' as the default. Please configure this value if this is wrong.
NVMain: the address mapping order is
Sub-Array 1
Row 6
Column 2
Bank 4
Rank 5
Channel 3
defaultMemory.channel0.FRFCFS-WQF capacity is 4096 MB.
Creating 4 command queues.
**** REAL SIMULATION ****
info: Entering event queue @ 0. Starting simulation...
Hello world!
Exiting @ tick 77407500 because exiting with last active thread context
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.subArrayEnergy 4.33674nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.activeEnergy 4.1412nJ
```

Hello world!

OUTPUT

- You can check the number of cache hits from stat.txt in gem5/m5out

The screenshot shows a terminal window with the file `stats.txt` open. The file contains a large number of statistics related to the system's performance, including cache hits, misses, and latency. The statistics are organized into sections, with some values being `nan` (not a number) or `0`. The file is located in the directory `~/gem5-525ce650e1a5bbe71c39d4b15598dec003cc9f9e/m5out`. The terminal window also shows a file explorer on the right side, with the file `stats.txt` selected. The file explorer shows the contents of the directory `gem5-525ce650e1a5bbe71c39d4b15598dec003cc9f9e`, which includes files `config.ini`, `config.json`, and `stats.txt`. The file `stats.txt` is highlighted, and a tooltip indicates it is 64.8 kB in size.

```
system.cpu.dcache.blocked_cycles::no_mshr 0 # number of cycles access was blocked
system.cpu.dcache.blocked_cycles::no_targets 0 # number of cycles access was blocked
system.cpu.dcache.blocked::no_mshr 0 # number of cycles access was blocked
system.cpu.dcache.blocked::no_targets 0 # number of cycles access was blocked
system.cpu.dcache.avg_blocked_cycles::no_mshr nan # average number of cycles each access was blocked
system.cpu.dcache.avg_blocked_cycles::no_targets nan # average number of cycles each access was blocked
system.cpu.dcache.ReadReq_mshr_misses::cpu.data 56 # number of ReadReq MSHR misses
system.cpu.dcache.ReadReq_mshr_misses::total 56 # number of ReadReq MSHR misses
system.cpu.dcache.WriteReq_mshr_misses::cpu.data 79 # number of WriteReq MSHR misses
system.cpu.dcache.WriteReq_mshr_misses::total 79 # number of WriteReq MSHR misses
system.cpu.dcache.demand_mshr_misses::cpu.data 135 # number of demand (read+write) MSHR misses
system.cpu.dcache.demand_mshr_misses::total 135 # number of demand (read+write) MSHR misses
system.cpu.dcache.overall_mshr_misses::cpu.data 135 # number of overall MSHR misses
system.cpu.dcache.overall_mshr_misses::total 135 # number of overall MSHR misses
system.cpu.dcache.ReadReq_mshr_miss_latency::cpu.data 14411000 # number of ReadReq MSHR miss cycles
system.cpu.dcache.ReadReq_mshr_miss_latency::total 14411000 # number of ReadReq MSHR miss cycles
system.cpu.dcache.WriteReq_mshr_miss_latency::cpu.data 12519000 # number of WriteReq MSHR miss cycles
system.cpu.dcache.WriteReq_mshr_miss_latency::total 12519000 # number of WriteReq MSHR miss cycles
system.cpu.dcache.demand_mshr_miss_latency::cpu.data 26930000 # number of demand (read+write) MSHR miss cycles
system.cpu.dcache.demand_mshr_miss_latency::total 26930000 # number of demand (read+write) MSHR miss cycles
system.cpu.dcache.overall_mshr_miss_latency::cpu.data 26930000 # number of overall MSHR miss cycles
system.cpu.dcache.overall_mshr_miss_latency::total 26930000 # number of overall MSHR miss cycles
system.cpu.dcache.ReadReq_mshr_miss_rate::cpu.data 0.051661 # mshr miss rate for ReadReq accesses
system.cpu.dcache.ReadReq_mshr_miss_rate::total 0.051661 # mshr miss rate for ReadReq accesses
system.cpu.dcache.WriteReq_mshr_miss_rate::cpu.data 0.083953 # mshr miss rate for WriteReq accesses
system.cpu.dcache.WriteReq_mshr_miss_rate::total 0.083953 # mshr miss rate for WriteReq accesses
system.cpu.dcache.demand_mshr_miss_rate::cpu.data 0.066667 # mshr miss rate for demand accesses
system.cpu.dcache.demand_mshr_miss_rate::total 0.066667 # mshr miss rate for demand accesses
system.cpu.dcache.overall_mshr_miss_rate::cpu.data 0.066667 # mshr miss rate for overall accesses
system.cpu.dcache.overall_mshr_miss_rate::total 0.066667 # mshr miss rate for overall accesses
system.cpu.dcache.ReadReq_avg_mshr_miss_latency::cpu.data 257339.285714 # average ReadReq mshr miss latency
system.cpu.dcache.ReadReq_avg_mshr_miss_latency::total 257339.285714 # average ReadReq mshr miss latency
system.cpu.dcache.WriteReq_avg_mshr_miss_latency::cpu.data 158468.354430 # average WriteReq mshr miss latency
system.cpu.dcache.WriteReq_avg_mshr_miss_latency::total 158468.354430 # average WriteReq mshr miss latency
system.cpu.dcache.demand_avg_mshr_miss_latency::cpu.data 199481.481481 # average overall mshr miss latency
system.cpu.dcache.demand_avg_mshr_miss_latency::total 199481.481481 # average overall mshr miss latency
system.cpu.dcache.overall_avg_mshr_miss_latency::cpu.data 199481.481481 # average overall mshr miss latency
system.cpu.dcache.overall_avg_mshr_miss_latency::total 199481.481481 # average overall mshr miss latency
system.cpu.dcache.replacements 0 # number of replacements
system.cpu.dtb_walker_cache.tags.pwrStateResidencyTicks::UNDEFINED 77407500 # Cumulative time (in ticks) in various power states
```


ACTIVE ENERGY

- After running the program, you can see it in terminal.

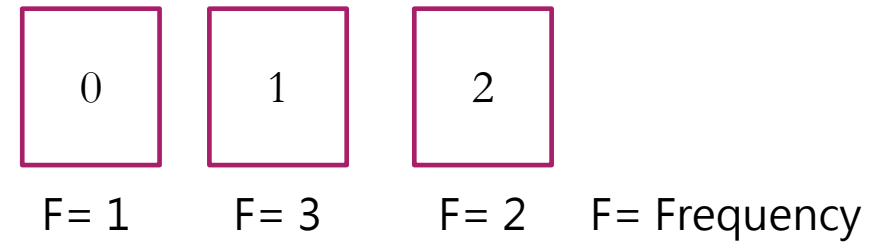
```
test@ubuntu: ~/gem5-325ce650e1a5bbe71c39d4b15598d6c003cc9f9e
File Edit View Search Terminal Help
Channel 3
defaultMemory.channel0.FRFCFS-WQF capacity is 4096 MB.
Creating 4 command queues.
**** REAL SIMULATION ****
info: Entering event queue @ 0. Starting simulation...
Hello world!
Exiting @ tick 77407500 because exiting with last active thread context
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.subArrayEnergy 4.33674nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.activeEnergy 4.1412nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.burstEnergy 0.195536nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.writeEnergy 0nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.refreshEnergy 0nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.cancelledWrites 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.cancelledWriteTime 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.pausedWrites 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averagePausesPerRequest 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.measuredPauses 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averagePausedRequestProgress 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.measuredProgresses 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.reads 121
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.activates 51
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.precharges 50
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.refreshes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.worstCaseEndurance 1844674407370955161
5
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averageEndurance 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.actWaits 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.actWaitTotal 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.actWaitAverage -nan
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.worstCaseWrite 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num0Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num0Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num10Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num11Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averageWriteTime 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.measuredWriteTimes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.mlcTimingHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.cancelCountHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.wpPauseHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.wpCancelHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.bankEnergy 4.33674nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.activeEnergy 4.1412nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.burstEnergy 0.195536nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.refreshEnergy 0nJ
```

ENABLE L3 LAST LEVEL CACHE

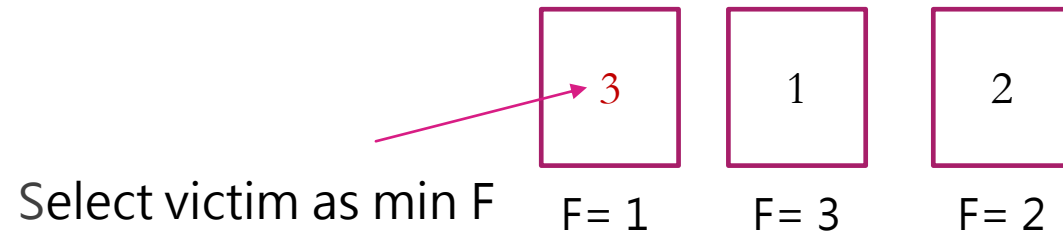
- The files that need to be modified are in the gem5 folder
 - Options.py
 - Caches.py
 - Xbar.py
 - BaseCPU.py
 - CacheConfig.py
- The first four files only increase the parameters of the L3 cache, and you can imitate them according to the settings of the L2 cache.
- CacheConfig.py needs to connect the L3 cache to the entire Gem5 system. Pay attention to the relationship between the L2 and L3 caches. The system can only use the L3 cache when the L2 cache is already used. This condition is not met.
- For the details of the code, students can go online to find resource keywords like Gem5 L3 cache.

FREQUENCY BASED REPLACEMENT POLICY

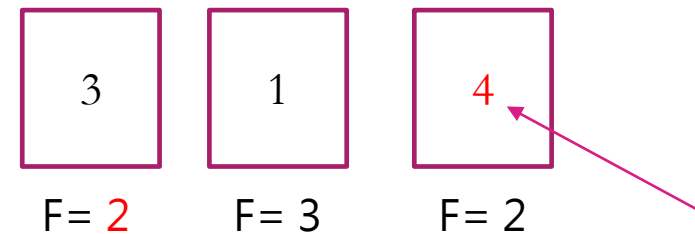
- Cache size = 3 full associative blocks
- block request 0, 1, 2, 1, 2, 1



- block request 3



- block request 3, 4



Select victim as min F
and the Oldest data.

WRITE BACK AND WRITE THROUGH

- You need to understand write back and write through mechanisms.
- You can go to the mem/cache folder to find two files, one is called cache.cc and the other is called base.cc. You can study how the underlying code of gem5 writes data into Memory.